X Sentiment Analysis

Apple & Google Products

Automatically classifying customer sentiment from tweets to support real-time brand monitoring

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The Problem

Not Scalable

Thousands of tweets arrive daily. Manual review is impossible.

Generic Tools

Existing tools are expensive and not trained on tech brand language.

Slow Detection

Negative sentiment goes undetected until it becomes a crisis.

We need an automated way to know what customers are saying — instantly.

Our Solution

A machine learning model that reads a tweet and automatically classifies it as:

😊 **POSITIVE**

"Love the new iPhone!"

😠 **NEGATIVE**

"Google Maps crashed again!"

Built using 9,093 real tweets about Apple & Google products

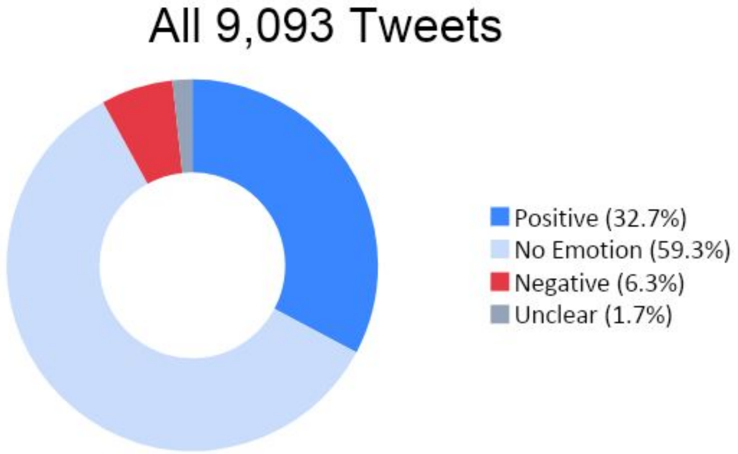
The Data

9,093
Total Tweets

3,548
Used for Model

84%
Positive Tweets

16%
Negative Tweets



Key Challenge:

More positive than negative tweets.
Model trained to handle this imbalance
so it catches negative tweets reliably.

How It Works

1. **Raw Tweet**
   Customer posts a tweet about Apple or Google

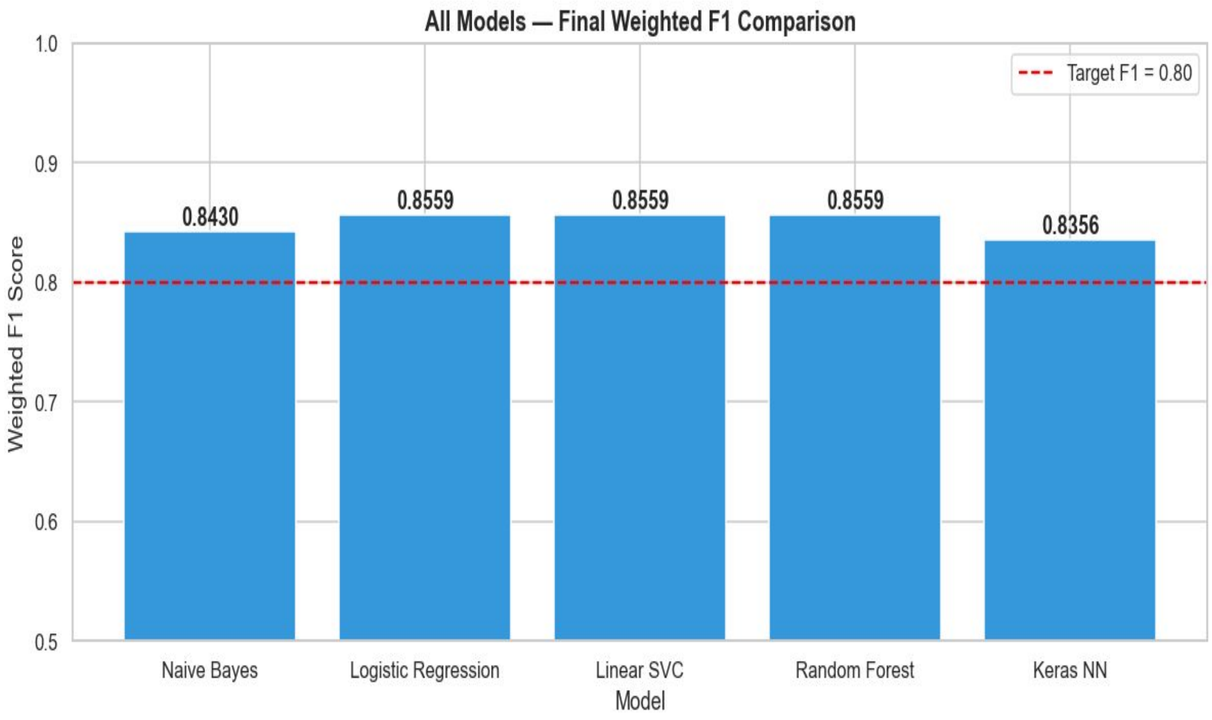
2. **Clean Text**
   Remove links, symbols, noise and stopwords

3. **Convert to Numbers**
   TF-IDF gives each word a weighted score

4. **Model Predicts**
   Logistic Regression classifies the tweet

5. **Result**
   😊 Positive or 😠 Negative

Model Performance



Success Metrics

Weighted F1
0.8559
✅ Met

ROC-AUC
0.8624
✅ Met

CV vs Test F1
0.0160
✅ Met

Neg. Recall
0.4800
❌ Not Met

🏆 **Best Model: Tuned Logistic Regression**

What the Model Found

Product Insights

iPad attracted the most positive tweets (793)

iPhone had the highest share of negative tweets

Apple and Google brand mentions are the most frequent

Key Words Driving Predictions

😊 Positive Words

love · great · free · awesome · new · store

😠 Negative Words

fail · crash · hate · broken · headache

💡 The model explains WHY it made each decision using SHAP & LIME interpretability tools — it is not a black box.

Recommendations & Next Steps

🚀 Deploy in Real Time

Connect the model to a live Twitter feed for instant sentiment alerts.

📊 Live Dashboard

Pipe predictions into Power BI or Grafana for daily brand health tracking.

🔄 Expand to 3 Classes

Add neutral sentiment to cover the full range of customer expression.

📦 More Diverse Data

Collect tweets beyond SXSW for broader, year-round brand coverage.

Business Value: Respond faster · Track brand health daily · Make decisions backed by customer voice

Thank You

Questions & Discussion

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